

CHECK WHAT YOU LEARNT

Check Yourself - Unit 2**A. Choose the correct answer**

- Code inside void setup() runs
(a) forever (b) once at the start (c) only on reset (d) never
- digitalWrite(9, HIGH) means
(a) read pin 9 (b) switch pin 9 on (c) switch pin 9 off (d) fade pin 9
- delay(500) makes the board wait for
(a) 500 seconds (b) 500 minutes (c) half a second (d) 5 seconds
- Before uploading you must always check
(a) the LED colour (b) board and port (c) the room light (d) the battery
- Sensors that give a value, not just ON or OFF, connect to
(a) D13 (b) GND (c) analog pins (d) the USB port
- Almost every instruction in Arduino code ends with
(a) a comma (b) a semicolon (c) a bracket (d) nothing

B. Fill in the blanks

- Code that must repeat forever is written inside void _____.
- A line beginning with // is called a _____.
- pinMode(2, _____) tells the board that pin 2 will listen.
- Pins marked with the ~ symbol can _____ an output smoothly.

C. Answer in one or two lines

- Explain the difference between setup() and loop() in your own words.
- Your code uploads successfully but the LED stays dark. Is the problem in the code or the wiring? How would you find out?
- Why do programmers write int led = 13; instead of typing 13 everywhere?
- Write the one line of code that switches pin 7 off.

UNIT 3

Basic Project: Traffic Light Signal Controller

Sessions 6, 7 and 8

Your first real machine. A traffic signal is nothing more than three lamps switched on in a fixed order with the right waiting time between them. Simple - and yet it keeps millions of people alive every single day.

By the end of this unit you will be able to:

- Explain the order and the purpose of red, yellow and green.
- Draw a flowchart before you write any code.
- Connect a four-pin module to the correct digital pins.
- Test your own machine and improve the timing after watching it run.

SESSION 6

one class

The Problem and the Plan

Today you will

- Study how a real traffic signal behaves.
- Decide the timing for red, green and yellow.
- Draw the flowchart for your machine before touching a wire.

Outside your school gate, students and vehicles share the same stretch of road. Nobody knows whose turn it is, so everybody edges forward together. Your job is to build a signal that repeats a safe cycle entirely by itself, with no human pressing anything.

Before an engineer touches a wire, they draw the plan. A **flowchart** is a picture of what a machine will do, step by step, in the order it will do it. It costs five minutes and saves an hour. Here is the cycle you are going to build.

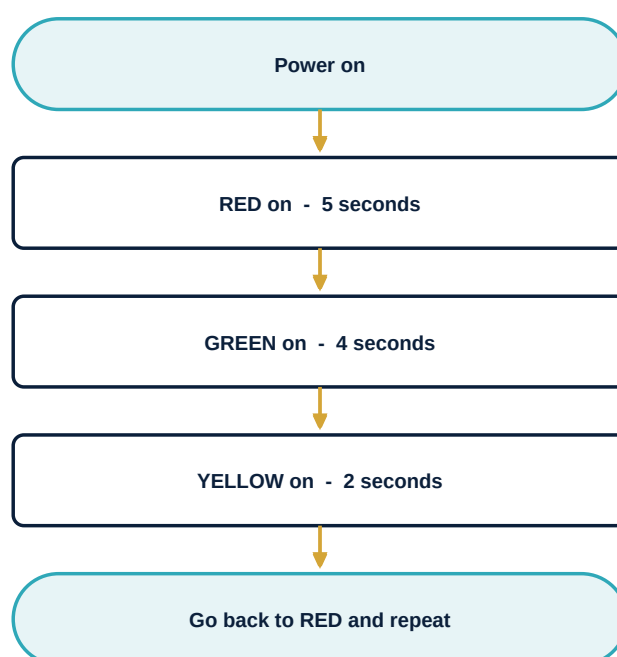


Figure 3.1 - The traffic cycle, drawn before it is coded

Why is yellow needed at all? Because a driver travelling at speed cannot stop instantly. Yellow is a warning that green is ending, giving time to slow down safely. Remove yellow from a real junction and you get collisions. Every part of a machine exists for a reason - always ask what the reason is.

Try This

Before the next class, stand safely near a real signal for two minutes with a watch. Time each lamp. Write the three numbers in your notebook and compare them with the numbers in the flowchart above. Are yours different? Why might that be?

SESSION 7

one class

Wiring and Coding the Signal

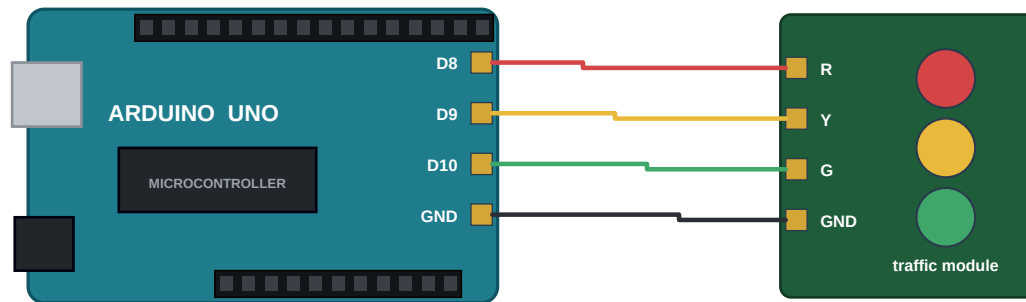
Today you will

- Connect the traffic light module to the correct pins.
- Type, verify and upload the traffic light program.

- See your flowchart running as real hardware.

The traffic light module already has the three lamps *and* their resistors built onto it. That is why you need only four wires instead of eight: one for each lamp, plus one shared ground. This is exactly why modules exist - somebody has done the fiddly part for you.

Traffic light module -> Arduino Uno



Only four wires. Match your jumper colours to the lamps - it saves hours of debugging.

Figure 3.1 - Wiring the traffic light module

Module pin	Arduino pin	Why
R	D8	Red lamp
Y	D9	Yellow lamp
G	D10	Green lamp
GND	GND	The return path. Without this, nothing works at all.

Watch Out

GND is not optional and it is not decoration. Electricity must have a way back to the board or the loop is never completed. Roughly nine out of ten 'my module is dead' problems turn out to be a missing GND wire.

TrafficLight.ino

```
int red    = 8;
int yellow = 9;
int green  = 10;

void setup() {
  pinMode(red, OUTPUT);
  pinMode(yellow, OUTPUT);
  pinMode(green, OUTPUT);
}

void loop() {
  digitalWrite(red, HIGH);    // STOP
  delay(5000);
  digitalWrite(red, LOW);

  digitalWrite(green, HIGH); // GO
  delay(4000);
  digitalWrite(green, LOW);

  digitalWrite(yellow, HIGH); // GET READY
  delay(2000);
  digitalWrite(yellow, LOW);
}
```

Read the code and the flowchart side by side. They say exactly the same thing - one in pictures for you, one in words the board understands. Notice also that each lamp is switched off before the next one comes on. Delete one of those LOW lines and see what happens; it is a mistake worth making once.

SESSION 8

one class

Testing, Tuning and Caution Mode

Today you will

- Test your signal the way an engineer would.
- Change the timings and justify your choice.
- Add a night-time caution mode as a challenge.

A machine that worked once is not a machine that works. Engineers test deliberately, write down what they see, and only then say the job is done. Work through this table with your group and tick each row honestly.

Test	What should happen	Tick
Power on and watch three full cycles	The same order every time, nothing skipped	■
Count the seconds with a phone timer	Matches the delay values you wrote	■
Wiggle the jumper wires gently	Lamps stay steady, nothing flickers	■
Pull out the GND wire	Everything stops - proof that it matters	■
Ask a friend to watch one cycle	They can correctly predict the next lamp	■
Leave it running for five minutes	No lamp gets warm, nothing resets by itself	■

Now tune it. Watch your own signal for a while and ask whether the timings feel right for a school gate, where people walk rather than drive. Should red be longer? Should yellow be shorter? Change the numbers, upload again, and write down **why** you chose what you chose. An engineer who cannot explain their numbers has only been guessing.

Challenge

Real signals switch to **caution mode** late at night: the yellow lamp alone blinks on and off, once per second, and the other lamps stay dark. Add a caution mode to your code. Then think about how you would let the machine choose between normal mode and caution mode by itself - you will be able to solve that properly after Unit 4.

Did You Know

The order red - green - yellow is not the same everywhere in the world. In several countries the signal shows red and yellow together just before green, warning drivers to get ready. Your code could do that with a single extra digitalWrite.

New word	What it means
Flowchart	A picture of what a machine will do, step by step.
Module	A small ready-made board with parts already fitted on it.
Cycle	One complete run through a repeating sequence.